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Contrasting Simulations and Experiments in Social Sciences: Materiality Thesis and Epistemic Privilege Revisited

Abstract

This dissertation offers a social science scope to contrast simulations and experiments. The comparison of simulations and experiments starts with identifying the ontological gap and targets at acknowledging the alleged epistemic privilege. The Materiality Thesis is commonly thought to be the most influential account to go through the above process. The material difference exists on two levels: simulating systems and experiment systems, as well as the relationship between simulating systems or experimental systems and the target system. I argue that the Materiality Thesis is inarticulate and outdated in differentiating simulations and experiments in social sciences. What really sets simulations and experiments apart is the different ways to achieve relevant similarity with the target system. The argument of epistemic privilege thus should be revisited. I provide a pragmatic perspective and argue that neither simulations nor experiments entitle epistemic privilege.

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1 INTRODUCTION

There is a tendency in the scientific community to see simulations work as a distinct method in scientific studies. There are abundant studies in social sciences using simulation as a unique methodology to investigate phenomena of interest (e.g., Schelling, 1971; Hegselmann, 2002; Giabbanelli, 2013; Dykstra et al., 2013). The interesting as well as pressing question for scientists and philosophers to answer is: what is the distinctive characteristic of simulation that makes it a unique method? In this dissertation, I adopt the most common discussion scope, i.e., contrasting simulations and experiments. To see examples of simulations and experiments in social sciences studies:

> Case of Simulation: To study the opinion dynamics and social influences, a simulation is constructed based on agents' behaviour with reference to the Theory of Reasoned Action and Social Judgement Theory. The agents in the simulation gamble on debates, seeing the various impacts of the outcome of debates on the reputation of the agents. (See Dykstra, 2013)
>
> Case of Experiment: To determine if workers will become more productive in higher or lower light levels. The experiment is designed with controls on the environment, isolating two groups of workers and changing only the light levels in one group to see the variation tendency in both groups of workers' daily output. (See McCarney et al., 2007)

At first glance, they are very dissimilar methods in scientific applications. Yet there are some affinities between simulations and experiments that vindicate the scope of contrasting is appropriate. First, both simulations and experiments need to set up the initial environment and manage input in the hope of obtaining informative output through the process of simulation and experimentation. Second, simulations and experiments share the general wish of learning about the part of the world of interest that they represent or replicate (Beisbart, 2018; Jebeile, 2019). In other words, simulations and experiments are similar in the aspects that controlling behaviours are both crucial, the foci on the unknown outputs are equivalently strong, and the hopes of extrapolating conclusions to the target system of interest in the real world are alike.

What are simulations and experiments, and how do they differ from one another, then? Much talk about the comparison of simulations and experiments suggests an investigation into their ontological properties. The most influential account of the ontological difference is the Materiality Thesis. There are two levels of analysis in the Materiality Thesis. The first level of analysis focuses on the material difference between the simulating and experimental systems, assessing to what degree the material stuff engages in simulating and experimental entities. The next level of analysis focuses on the difference in the material relationship of simulating and experimental systems to the target system of interest, claiming that experiments link to the target system by material relationships, whereas simulations only obtain formal relationships.

I argue that the focus on the material status of the simulating and experimenting system is misplaced. The meaningful difference between simulations and experiments shall derive from the analysis of the way they are constructed, i.e., how they relate to the target system. In this way, the meaningful characteristics of the simulating and experimental systems also reveal epistemically. I argue that simulations and experiments both aim at achieving a relevant similarity with the target system but in different ways. Simulations are constructed based on good modelling, whereas experiments are built based on kindhood recognition.

Given the difference between simulations and experiments lies in the ways to reach relative similarity, the ground of the discussion about epistemic privilege thus shifts from materiality to constructive characteristics. Commonly, the criterion of ascertaining epistemic privilege is developed around the concept of knowledge. This view focuses on the knowledge itself and treats simulations or experiments as ways to touch the boundaries of human thoughts. I offer a pragmatic perspective to discuss epistemic privilege. The pragmatic perspective focuses on inquiry in addition to knowledge. I argue that neither simulations nor experiments entitle epistemic privilege since both can produce special and useful knowledge according to the inquiry and that knowledge cannot be obtained by another method.

This dissertation will be structured as follows. In Section 2, I will introduce the conceptual works about simulations and experiments and summaries to give a comprehensive picture of their work in social sciences. In Section 3, I will present the two levels of analysis in the Materiality Thesis. I will also remark on the criticisms and consider a more suitable framework for the differentiation account in social sciences. In Section 4, I will explain how to read and answer the problem of epistemic privilege. I will use a pragmatic perspective to show how simulations and experiments produce special knowledge and why the other onemea cannot obtain it. After that, I summarise my arguments and conclude in Section 5.

2 ON THE DEFINITIONS OF SIMULATION AND EXPERIMENT

2.1 The Definition Work of Simulation

Citing various definitions of the word "simulation" in dictionaries, simulations can be broadly understood as an imitation of a situation or process. This is the literal meaning of the word "simulation" but also a candidate for the definition projected by philosophers. For example, Hartmann (1996) defined simulation as follows:

> A simulation imitates one process by another process. (p.83)

This definition is broad in a way that any process working as a proxy to represent another process can be seen as a simulation. A lot of practices and activities in daily lives thus fall into this very

basic definition, for example, reality-based games like Plague Inc. or Monopoly, reality-enacting activities like a fire drill or an interview rehearsal. This definition definitely captures important truths about simulation, yet not accurate enough for the analysis, particularly in scientific practices.

To be more specific and accurate about the work simulations do in social sciences, the extensive use of computers as implementing tools should be introduced. In most cases, scientists do not obtain social knowledge by analogous simulation, instead, they run computer simulations to gain useful solutions of designed models that say something about the social phenomena of interest. There are good reasons that simulations in social sciences are often computer simulations. Speaking from a methodological point of view, only computer simulations are capable of processing parameters and models numerically. There are also pragmatic reasons like significant cost reduction, freedom from ethical concerns, etc., supporting the development of computer simulations in social sciences (Reiss, 2011).

There are several attempts to define and characterise simulations in the philosophical literature that forms a long-lasting debate (e.g., Humphreys, 1991; Hartmann, 1996; Guala, 2002; Humphreys, 2004; Winsberg, 2010, etc.). In what follows, I attempt to provide a structure of these definition works of simulations. Constantly, a choice or a combination of the following parts helps us to understand a simulation practice in scientific practices.

> Descriptive Part: Simulation is an imitation of a process (Hartmann, 1996, p.83).
>
> Technical Part: Simulation is a process of building, using, and justifying a computational model which entails analytically intractable mathematics (e.g., Humphreys, 1991, p.501; Winsberg, 2001, p.443; Reiss, 2011, p.244; Arnold&Durán, 2013, p.40, etc.).
>
> Characteristic Part: Simulation is an exploration of computational models which aim at drawing inferences about the social system of interest (Reiss, 2011, p.245).
>
> Purpose Part: Simulation is an approach that enables the study of a complicated system which is difficult or impossible to study in the first place.

The definitions provide different parts of characterisations of simulations. These definition works are all to some extent patchy respectively, yet together elucidate simulations well. To explain more on this point, I have argued that the descriptive definition is too broad to involve many irrelevant cases in the discussion of simulations in social sciences. As for the technical part, it is indeed the very foundation and beginning of the simulating technique, particularly as a means to separate analogous simulations and set a neat discussion scope for computer simulations exclusively. It is the capacity of performing iterative computation of various kinds of equations to offer numerical rather than analytical solutions that link simulations with computers as necessary realisation tools. Yet there are many acknowledged simulations in social sciences practices that do not emphasise that much on the technical part of advanced solutions of equations. For instance, the agent-based

modelling as one method in social simulation, the most outstanding characteristic is not the overarching mathematical model awaiting numerical solutions but some simple local rules directing the interaction of the individuals comprising the simulation (Grüne-Yanoff, 2011). To motivate other interesting discussions such as materiality status and epistemological power, further characterisations of simulations are needed for the supplement. The characteristic part and the purpose part of the definition together correctly draw a typical simulation in social sciences studies, which targets at the inference to the real world and moderate the investigation conditions.

2.2 The Definition Work of Experiment

Crudely put, an experiment in social sciences is a research strategy to investigate and establish the influence of one variable while other variables are controlled. This definition can be read as a combination of two crucial conditions: experimental setting and research intention. The experimental setting involves identifiable intervention or manipulation and a controlled condition[1]. The conditions to form an experiment can be listed as follows:

1. Intervention;

2. Controlled setting.

So far it is insufficient to say an activity which involves intervention and a controlled setting is an experiment. One would not say that the mere phenomenon of pedestrians crossing the road after the green light is on is an experiment. The reason for this disqualification is the absence of a conscious observation or investigation. The researcher's intention on obtaining empirical knowledge is necessary otherwise it will render the concept of the experiment too broad to be useful. It is also noteworthy that the intervention and observation should be applied to the same object. Otherwise, the way an experiment reaches its conclusion would not be legitimate. Therefore, two more conditions should be added:

3. Observation;

4. Intervention and observation are posed on the same object.

Compared to the definition works of simulations, what is of the experiments' core is clear and uncontroversial: intervention and observation. The question is how to set the boundary of the scope of experiments because there are all sorts of scientific activities that fall into this category. To make the discussion of the differentiation of simulations and experiments clear, I would mainly consider laboratory/conventional experiments as prototypes.

[1] There is a wide range of activities that do not contain a controlling environment, yet they are given the labels of "experiment", e.g., Aristotle's observation of the chick (Durán, 2018). Here I do not wish to get into the debate of the overlapping in observation and experiment. Disagreements on this point shall not matter too much for the arguments to follow.

2.3 Overlaps in Concepts and Practices

So far, I have presented the definition works of simulations and experiments respectively. The definition works should resonate with most of the impressions about simulating and experimental activities in practices, albeit very broadly. As a result, one will notice that two broad pictures of simulations and experiments can have overlaps since the definition works are not designed in a mutually exclusive way. One can conduct a scientific activity that is designed as researchers observing the influence of an intervention in a controlled setting, meanwhile working as a representation of another system of interest. In other words, according to the current definition works, we can build a simulation that looks like an experiment also an experiment having simulating characteristics.

In fact, this kind of scientific activity is what one finds most in social sciences studies. For example, in a research about the social impact of whistleblowing (Xiu, et al., 2001), the simulating and experimental setting can be easily spotted and recognised. This research is designed to study the dynamic process of an organisation's changing ethical environment (e.g., organisational injustices) by applying social simulation. The researcher particularly finds out the effectiveness of whistleblowing to prevent organisational injustices. In this case, the changing organisational ethical environment is simulated, and the factor of interest – the presence of whistle-blowers works as an intervention and is observed. Starting from the definition, this research is both practices of simulation and experiment. In other words, because the definitions are not mutually exclusive, it seems that scientific activity can be a simulation and an experiment simultaneously as long as a dynamic process involves an identifiable intervention. This is exactly the most common case we see in computational social sciences practices, that we typically do not identify a study as a simulation or an experiment but find the entangled simulating and experimental practices in it explicitly.

3 Contrasting Simulation and Experiment

The above definition works of simulations and experiments give us a rough sense of what a simulation and an experiment look like in social sciences. Given the grey areas in both concepts and practices, a pressing problem raises: Is it possible to find a neat division between simulations and experiments? If it is, then what is the decisive factor to set simulations and experiments apart? As I have stated before, scientists and philosophers of science pay attention to this comparison of simulations and experiments because they are interested in whether they serve as the same means to produce an understanding of the world. To investigate the epistemic entitlements of simulations and experiments, they try to start from the ontological ground. The Materiality Thesis is one of the attempts.

The attention to simulations and experiments' material status is deeply rooted in scientific intuition as well as historical discussion. For instance, Gilbert and Troitzsch (1999) contend that one controls for the actual thing of interest in a real experiment, whereas in a simulation one is experimenting with a model rather than the actual object of interest (Grüne-Yanoff & Weirich, 2020). There is a rough sense that simulations are virtual while experiments are concrete, simulations are "fake" while experiments are "real". Albeit very roughly, this sense does follow the rationale of the Materiality Thesis.

3.1 Two Levels in Materiality Thesis

In this section, I frame the Materiality Thesis in two levels of claims. The first level of analysis is about the properties of the simulating or the experimental entities. The second level of analysis is about the relationship between, on the one hand, the simulating system and on the other hand, the experimental system to the target system.

3.2 The Difference between Simulating and Experimental Systems

The first level is the material difference between simulating and experimental systems:

> [...] "virtual" experiments, ones in which we have nonmaterial experiments but which may involve some kind of mimicking of material objects. [...] It consists of statistical or mathematical models that are simulated, or "run", to generate output series with the aim of mimicking observed economic time-series data. (Morgan, 2003, p.217-225).

Here, a virtual experiment can be roughly understood as a characterisation of computer simulation, or at least an experiment that involves a very high degree of designs and settings of mathematical models. To reiterate this argument in the structure of the Materiality Thesis, see Parker's (2008b) sketch:

> To some extent, however, her characterisation of computer simulation studies as nonmaterial experiments seems to depend on their being understood as experiments on mathematical models or mathematical objects rather than on real, physical stuff (p.488).

Briefly speaking, here computer simulations are abstract entities, whereas experiments are actual, concrete entities. The difference between simulations and experiments thus lies in the material status of the simulating and experimental systems.

Simulations are constructed by mathematical models that can be read and function on a computer. It should correctly construe the essential descriptions and specifications of the objects in the target system regarding the purpose of various studies. For example, researchers create models of digital individuals and artificial communities based on theories which represent real-world individuals' behaviour with sufficient exactitude. For example, Social Judgement Theory is translated into

mathematical expressions to be the rules for the agents in the simulation to act correspondently. The process to know how the outcome of debates would impact agents' reputation and the dynamics of opinion is an intensive computation of models to a numerical solution.

Experiments, especially laboratory experiments, are composed by actual material which is correspondent with the real-world systems. Through processes such as sampling can researchers find materially analogous or even the same objects to study. Compared to the opinion simulation, the experimental intervention involves an action upon or the creation of a material object, i.e., real people in society. One could pose different results of debates on real participants. To see what are the influences of the results of debates and how opinions in a community flow is a natural reaction.

In this section, I argue that the simulations are nonmaterial entities, however, experiments are not purely "material" entities due to the constrained reading from a physical perspective. Thus, the Materiality Thesis's first level distinction fails to differentiate between simulations and experiments.

3.2.1 Parker's Objection: The Material Status of Simulations

Parker's objection points out an apparent mistake in the mindset of thinking simulation as a nonmaterial thing. It indicates the material status of the simulation, namely, the programmed computer. The core idea in this objection is that a simulation involves actual interventions on a material system, i.e., the computer as a physical being made of wire, plastic, silicon chips and so on. Every time researchers manipulate the mathematical models, they will have a concrete object to work on. It is a very straightforward illustration of the material status of simulation.

Moreover, Parker appealed to the role that a material computer plays in forming a simulation to reinforce her argument of the emphasis on the material status of simulations. The participation of a physical computer has to be counted in the simulating system because if not, the dynamics in the simulation would not have been formed. That is to say, every changing step in the mathematical system has to be explained by the brute computing force provided by programmed computers. Recall the definitions of simulation from the mathematical perspective, scientists undertake simulations when there are no handy options to solve equations analytically. Computers reach the numerical solution through billions of computational steps in sophisticated simulations. Material computers sponsor the computing power, so it is not just a realisation tool for simulating systems, but an essential constituent of it that ensures its very fundamental functions and characteristics of simulation.

3.2.2 Defense: A Possible Division

I do not think this objection works. The physical computer and the simulation can be divided as tools for the realisation of the representation and the representation itself. Only the representation is what we call a simulation. The reliance on the realisation of such representation does not indicate

an inseparable account of seeing the device and the output as a whole, even if it is necessary for pushing the iteration of status. Our thoughts are generated by the brain, and we would not have thoughts without the existence of the brain, but this does not mean that our thoughts are not abstract but material entities. Also, it is too demanding to insist that the physical existence of the tool for us to realise equations, models and calculations should be taken into account when talking about the properties of working entities. In this way, there will be no nonmaterial concepts, calculations, models and so on, for they all need some sort of material vehicle to do so.

Moreover, simply pointing out the material device is not enough to claim that a computer simulation is a material simulation. The fatal flaw in the claim of a material simulation is its incompetence to link the series of states during the computer output with physical changes. It is similar to that we cannot obtain precise thoughts and ideas by recognising the physical activities and states in our brains and nervous systems. This is the so-called epistemic opacity that there is no possibility that one can access the internal states of a physical computer and assign simulating meaning to it (Humphrey, 2004). To explain more, the different locations inside a computer, e.g., the memory, the processors, and the electronic filters are cognitively inaccessible for unaided humans (Durán, 2018). One does not recognise what changes are taking place in the electronic filter, let alone understand what the changes mean. In a nutshell, a simulation must not be a concrete stuff without any cognitive access, but an explicable image or phenomena represented by that concrete stuff. Thus, the mere dynamic mathematical system qualifies as a simulation.

3.2.3 Objection: The Nonmaterial Status of Experiments

As the material statuses in simulating and experimental entities are the decisive factor in differentiating simulation and experiment, and I have handled Parker's objection and contradicted that simulation entities are nonmaterial (in a physical sense), then the experiments should be material (in a physical sense) to support the Materiality Thesis.

However, some of the experiments can be nonmaterial in a physical sense. Consider a type of unit of analysis in social sciences: social interactions. Instead of studying individual humans or human-made artifacts, one can study what goes on between them. For example, cultural customs, privacy protocols, laws and regulations, TV shows, etc. These objects in studies seem not "physical" or, to say, "nonmaterial". For example, suppose one wants to study whether the upgrade of suppression of free speech will affect people's willingness to speak in the public sphere. This is a very simple experimental setting with identifiable intervention and control which are clearly nonmaterial without actions taken on concrete, physical material.

This objection may seem like a literal understanding of the materiality of the experimental entities. A possible defence would admit that social interactions and that kinds of objects are nonmaterial in a physical sense but real and indeed material in a social sciences sense. In other words, social interaction is something that is felt and proven to exist conclusively in our daily lives. Moreover, they are of significance in social sciences' senses. What "materiality" means is not mere physical

tangibility or visibility. Yet this will render the materiality thesis itself dubious and fails to serve as a distinction account because one could not say a simulation is nonmaterial in a physical sense, whereas an experiment is material in social sciences' sense.

It is true that the Materiality Thesis fails at this level with such reading, but not in a way Parker anticipates that simulations are material but the experiment can be nonmaterial in the sense that we say simulations are material. This is partly because the wording of "materiality" is outdated and confused especially in social sciences. It is understandable when early-stage discussion of the ontological differences between simulations and experiments takes place in natural science scenarios where there is a tradition of engaging physical stuff in the studies.

3.2.4 Redirection: Learn the System through the Inquiry of Relationship

Admittedly, this dispute of material status in simulating and experimental entities is a result of the ambiguity around the wording "material". The physical perspective which is emphasised in Parker's reading and objection is indeed misleading and constrained. The aim of this section is not to say simulating and experimental entities are the same. The crux is the wish that the Materiality Thesis at this level of analysis to serve as a successful account of distinction will fall through.

However, I do not find it warranted or informative, but rather superficial to do more elaboration or refinement of the discussion about the material status of the mere appearance of simulating and experimental entities. As we are struggling using words like "material", "substantial", "physical", "real", etc., the material status of simulating and experimental entities does not tell us any more about simulation and experiment. For instance, the way simulation and experiments relate to the real world, the kind and the quality of knowledge that simulation and experiments would bring, etc. In fact, what is more crucial is how simulating and experimenting systems are constructed respectively. Then the different characterisation of simulating and experimental system and their relationship to the real world shall follow naturally and meaningfully.

I shall move on to talk about the constructive characteristics of simulating and experimental systems in the next section through the inquiry of how they are related to the target system.

3.3 The Difference between Simulating System to Target System and Experimental System to Target System

The second level of the Materiality Thesis discusses the difference between the relationship of a simulating or experimental system and the target system and related consideration of simulating or experimental systems' constructive and functional characteristics.

3.3.1 The Working Procedure of Simulations and Experiments

To make this illustration clear, one should first get to know what is the relationship between a simulating or an experimental system and the target system, i.e., the reason and the way they link together. Then the characteristics of such relationships shall follow with such analysis. Generally speaking, both simulations and experiments serve as tools for us to gain knowledge of the part of the world that interests us. When a simulation or an experiment is appropriately constructed, we believe 1) it is justified to construct the simulating and experimental system in certain ways to study the part of the world of interest and 2) it is justified to use the knowledge gained from the simulation and experiment to form correlative knowledge of the part of the world of interest.

These two beliefs correspond with the two linkages in simulation and experiment studies (R1&R2 and r1&r2, see figure below). To explain the two relationships in other words, firstly a simulating or an experimental system for studying the target systems should be constructed. Secondly, one should make use of that simulating or experimental system with the intention of learning something about the target system.

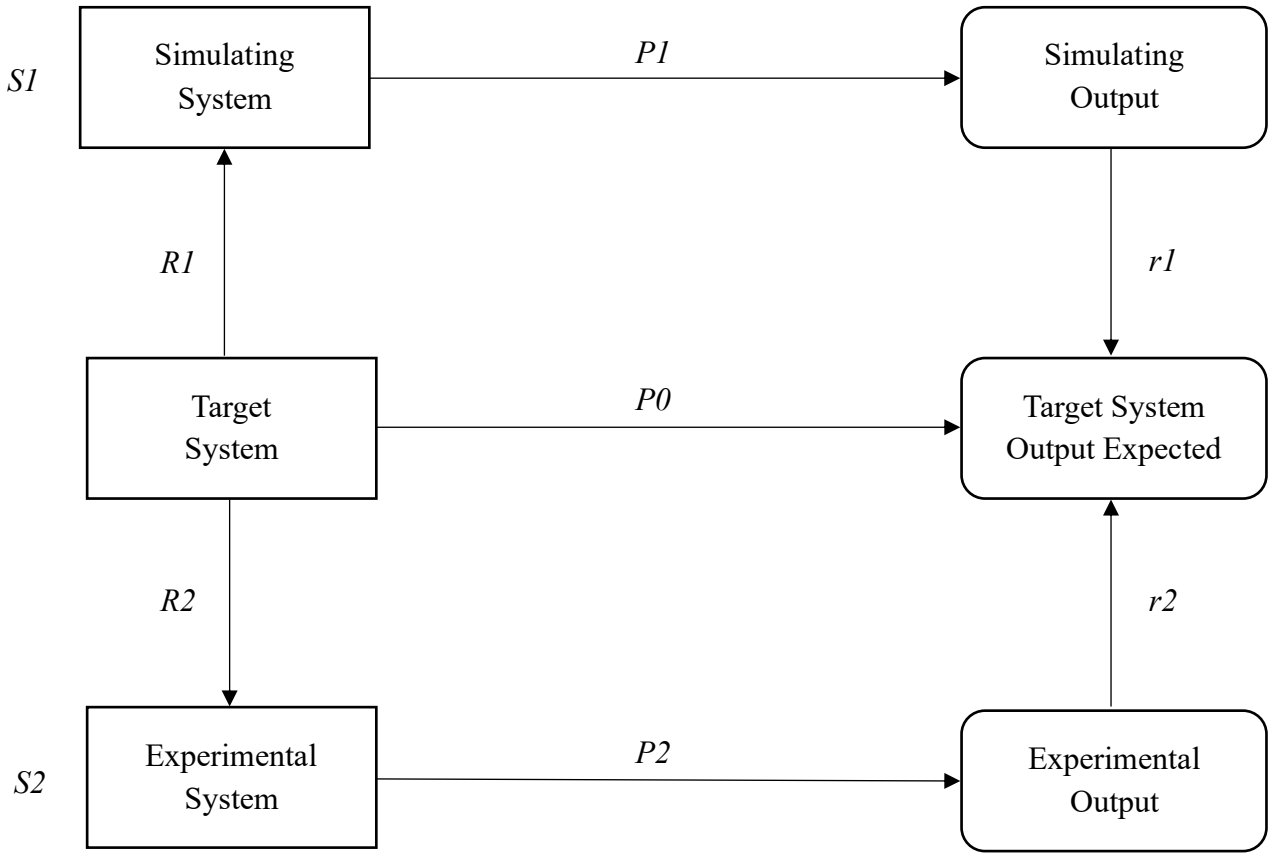

Figure. The Working Procedure of Simulations and Experiments

R: The relationship between the simulating/experimental system and the target system.

r: The relationship between the simulating/experimental output and the output of the target system.

S: The characterisation of the simulating/experimental system.

P: The way simulating/experimental output is formulated.

R1 and R2 are essentially the different ways a simulation and an experiment handle the reality of interest respectively. It is rather clear and uncontroversial that simulations must obtain such treatment for the target system. The way to do it is to abstract reality by using mathematical models as digital descriptions of the target system of interest as the definition works have illustrated. The trickier part is the way experiments handle the reality of interest.

In an ideal experiment, one could intervene and observe the exact target system. For example, if someone is interested in whether the different shapes of rubber affect its trajectory on a slope, he can knead a piece of rubber firstly into a square and secondly into a sphere and observe how the rubber moves in the different shapes. In this way, we say that experiments are conducted in reality without any additional treatment. One is directly intervening and observing that very object that he wants to investigate. However, this is not the most cases of experiments in social sciences. More often, scientists are not in a preferable position to study the very raw state of the target system because useful information is either hard to acquire or too poor to study (Guala, 2002; Durán, 2013). For example, if we want to study the organisational culture of a public company, we probably would not perform controls and interventions on that very public company as we wish, for possible reasons such as difficulties in environment control, the impermissibility of conducting the intervention, etc. Instead, we will either be limited in the scope of visiting that public company or have to find a substitute for that company which is easier or cheaper to study. It is hard to find many experiments without any construction of the raw reality and allow investigation of the object in a whole scope to ensure the exactness and completeness of the target system of interest. Most experiments more or less involve some sort of treatments to construct, e.g., sampling, substitution, etc., to handle the reality of interest. Certainly, different kinds of treatments retain various degrees of distance to the target system. For example, a good sampling may be the best way to construct an experiment with the best resolution and the least distance to the target system. The point is that experiments in social sciences are typically not the target system itself but need to perform treatment on the target system to be possible to study. Thus, the treatment and the justification of that treatment is the relationship between an experimental system and the target system.

Above is the first linkage starting from the target system and ending at a simulating or an experimental system, creating concessive versions of the target system of interest. The second linkage is from the simulating or experimental system to the target system. To be more specific, the linkage is the justification of the extrapolation regarding the simulating or experimental output

as well as the expected correspondence with the intended outcome of the target system. For example, if we can conclude from the lighting-productivity experiment that brighter lighting increases workers' productivity in the experimented factory, the next step is to assess whether it is permissible to say that changing the brightness in other factories will also have a similar effect on workers' efficiency. Also, if we get the output of a pandemic simulation which suggests that without further regulatory restrictions on people's social interactions, the infection rates will rise rapidly, what comes next is the consideration of whether we are justified in believing the real world will proceed as the simulation anticipates.

3.3.2 Guala's Materiality Thesis

Having explained the working procedure of simulations and experiments, let us see what the Materiality Thesis intend to argue about. See Guala's (2002) illustration of the difference in the relationship of simulating or experimental system and the target system:

> The difference lies in the kind of relationship existing between, on the one hand, an experimental and its target system, and, on the other, a simulating and its target system. In the former case, the correspondence holds at a 'deep', 'material' level, whereas in the latter the similarity is admittedly only 'abstract' and 'formal' [...].
>
> In a genuine experiment the same 'material' causes as those in the target system are at work; in a simulation they are not, and the correspondence relation (of similarity or analogy) is purely formal in character.
>
> The crucial presumption behind experiments is that relevant components of the laboratory system are made of the same 'stuff' as those of the target system. […] The experiment should feature the same causal processes that are at work in the real world, […] (Guala, 2002, p.66-67)

There are two arguments for the difference between the two kinds of relationships and constructive characterisations of the two systems. These claims are interrelated. The constructive characterisations of the simulating/experimental system (S1 and S2) indicate the relationship between the simulating/experimental system and the target system (R1 and R2). The former also ensures the way simulating/experimental output is formulated (P1 and P2), which speaks of the relationship between the simulating/experimental output and the output of the target system (r1 and r2).

System Characterisation

S1: The simulating system is made of different stuff from what makes the target system, presumably an abstraction of the stuff in the target system.

S2: The experimental system is made of the same stuff as what makes the target system.

Relationship Argument 1

R1: The relationship of the target system to the simulating system is abstract and formal.

R2: The relationship of the target system to the experimental system is deep and material.

Relationship Argument 2

P1: The simulating system produces output by formally corresponding relationships with what is in the target system.

P2: The experimental system produces output by the same material causal relationships as what is in the target system.

3.3.3 Objection: Shared Kindhood in Experiments

I shall handle the characterisations of the simulating/experimental systems (S1&S2) first and then argue against both relationship arguments. In this section, I deal with the Relationship Argument 1. What Guala demands in the experimental system is to contain the same objects made of the same stuff, just as what is in the target system. In this way, the material in the experiment can be seen as "a piece of the target system".

Despite the claim of "made of the same stuff" is very central to the Materiality Thesis, Guala has surprisingly little to say about how to recognise and assure that the experimental objects are made of the same stuff. One thing for sure is that he rejects a reductionist theory to say that every physical entity are made of the same fundamental sub-atomic particles (p. 67). That is to say, from Guala's perspective, the world should be capable of being categorised as "made by the same stuff" and "made by different stuff" with the target system neatly. Guala's example of the ripple tank is rather uncontroversial on this point as most cases in natural sciences. For instance, a ripple-tank model is built to study the wave nature of different kinds of waves. A ripple tank made of water tells us the nature and behavior of water waves. If one is interested in the water wave, then it is justified for him to study it in a ripple tank made of water to form knowledge. Because it is plausible to say that the sample water in the ripple tank is made of the same material as the water that he is originally interested in. Obviously, they are all aggregations of the water molecule (H_2O).

However, the argument of the experimental system is "made of the same stuff" as what is in the target system does not always work with clarity, especially in social sciences. Firstly, it is difficult to identify whether some non-physical objects are made of the same stuff. As I have argued in the previous Section 3.2.3 that, some of the objects in social sciences are not tangible or composed of some identifiable "stuff". For example, social interactions, organisations, etc. It is unclear how to read the "same stuff" and material similarity here.

Secondly and more importantly, for the experimental objects to be justified, the material similarity is neither necessary nor sufficient. What matters in the selection of experimental objects is a shared kindhood with the target objects (Roush, 2018). For example, suppose one is interested in whether the amount of the distribution of children's literature will impact children's character development. In this case, it is clear that the objects in the experimental system and the target system can be "made of the same material", they are both books of children's literature. One would not choose thriller books as experimental objects in this case because they are "made of different stuff". Consider another case that one is curious about whether the number of books in a city's collection is related to the city's cultural image. In this case, it seems that children's literature and thriller books can be treated as "being made of the same stuff" to contribute to the same object, i.e., the city's book collection. From these two cases, one could see the answers of whether the experimental objects and the target objects are made of the same stuff in the identical set varies according to the research.

In a very fundamental sense, children's literature is made of the same material as what makes thriller books, i.e., papers, glues, etc. Admittedly, this reading seems too implausible to proceed with. Then what do we have in mind when selecting children's literature but not thriller books in the previous case, but selecting both children's literature and thriller books in the case of cultural images in the latter case? Here, the research question, background knowledge, and causal conjectures are all at work. It is common sense that thriller books will have different ways to impact children's character development. Also, it is very clear that both the children's literature and thriller books will contribute to the amount of the book collection in an identical way, i.e., enumeration. This demonstrates that the experimental objects shall be of the same kind. The kindhood recognised may vary in different scenarios. As Roush correctly put it:

> Which properties we need to project the kind to depends on the question, and that projectability can co-exist with a great deal of difference in other properties.
>
> Material similarity is not sufficient for shared relevant kindhood. […] Material similarity is also not necessary (for shared relevant kindhood). (p.4900-4901)

The shared kindhood ensures that the experimental design catches the most relevant similarities with the target system in a researchable way. They are "adapted in the same kind" rather than "made of the same stuff". The rest of the characteristics out of the kindhood are insignificant to the experimental design thus ignorable. The material similarity is just a possible source of a shared relevant kindhood. Unlike natural sciences, the "made of the same stuff" and a start of materiality are themselves not so meaningful and mainstream for many social sciences interests. For instance, a study of one natural object's (e.g., a rock) chemical composition can be of importance, while such an investigation into a social object (e.g., an organisation) is less illuminating.

3.3.4 Objection: The Uncertainty of the Same Material Cause in Experiments

I have argued above that S2 and R2 are implausible and that the true relationship and characteristics are based on a shared relevant kindhood rather than similar/identical material. It should follow that the properties of the shared kindhood thus provide the relevantly similar procedure so as to produce the output and is justified to generalise to the target system (P1&P2 and r1&r2).

However, if the shared kindhood is only supported by a material similarity, such material similarity may still fail to replicate the exact material cause at work in the target system. For example, let us consider another major category of units of analysis in social sciences which clearly has a physical form - humans. Admittedly, in experiments investigating human behaviours, we typically conduct experiments on actual humans, which are, according to Guala (2002), "made of flesh, blood, neurons, etc. rather than silicon chips (p. 67)". Following this, all human beings appear to be made of the same stuff or share a natural kind. However, not all human beings produce the experimental output in an identical causal route. For example, two human beings can have very different decision processes, drug reactions, emotional changes, etc. That kind of different causal route is produced by the same kind of human beings or even organs which can appropriately be read as being made of the same stuff. All attempts to refute this objection and return to the material similarity claim have to reduce the unit of human beings to a lower level. Such reduction is doomed hopeless because we could not dismantle the material composition of humans and correspond to nonmaterial processes, e.g., mental states and drug reactions, etc. It is untenable to claim that two people who use separate decision-making processes are materially different from one another, especially when the material differences and corresponding causals are required to be articulated. As a result, the advocates of the material similarity and the same material cause in this case have to stop at a rather superficial level which may not produce many informative results.

To sum up, the experimental system is constructed by objects that share a kindhood with the objects in the target system. It is that shared kindhood ensures the experimental system and the target system are relevantly similar. Therefore, it is justified to generalise the observed output from the experimental system to the target system.

3.3.5 Confirmation of Simulating Relationship: Good Models

I believe Guala's claim about the relationship between the simulating system and the target system to be largely true, even though the relationship between the experimental system and the target system is in dispute.

I have argued for the nonmaterial status of the simulating system in Section 3.2.2. Explanations of the constructions of simulating systems are required to understand how this nonmaterial relationship got labelled as "formal" and "abstract". Simulations are built on the belief that we construct the models well, to put it bluntly. The simulation and the target system should be similar in most relevant facets through the construction of models.

The crucial point in how to build good models is that one already knows the specific characteristics of the target system that he is interested in learning about. In other words, the performance of the simulating system relies on the amount of background knowledge heavily. The background knowledge can be divided into two parts. One is to know what kind of rules are at work in the target system correctly. For example, in an agent-based model, it is important to design the models and parameters well to describe and reflect the behaviour of individuals in actual society. In other words, we have knowledge about what the world is like. This is an abstraction of the target system. Another kind of background knowledge is the techniques to build the mathematical models well. In other words, we have knowledge about how the world is represented in technical details. The abstraction of the target system is thus in a mathematical form.

To sum up, both simulations and experiments aim at achieving the relevant similarity with the target system but in different ways. Simulations achieve the relevant similarity by building good models whereas experiments recognise shared kindhood. Simulations produce output by formally corresponding relationships with what is in the target system. Experiments produce output by the shared kind of objects at work.

4 THE DEBATE OF EPISTEMIC PRIVILEGE: A PRAGMATIC PERSPECTIVE

In what follows the original discussion of the Materiality Thesis, advocates of such a view ground the investigation of the epistemic power on the identification of different material statuses and relationships. Alongside the Materiality Thesis, experiments have a material status and therefore secure a material relationship with the target system, which serves as the evidential source for a common view that experiments are epistemically privileged over simulations. As the Materiality Thesis itself is reviewed in the previous sections that the absolute material status and relationship are proved absent, the former focal point on the material conditions should be moved to somewhere else. I propose to follow the alternative of the differentiation account of simulations and experiments in this dissertation. Because it shows the constructive characteristics of experiments and simulations to achieve relevant similarities in an epistemically informative way.

4.1 The Criteria of Epistemic Privilege

Even though the Materiality Thesis is argued as outdated and inarticulate in social sciences, the works of assigning epistemic privilege still provide some references about the criteria. How do the differences between the ways of achieving relevant similarity lead to epistemic privilege? Or, what exactly makes simulations or experiments superior compared to the other one?

There are several accounts in reading epistemic privilege. Winsberg (2010) grants priority to experiments from a time sequential perspective (p.71). By this he means to build good simulations, one needs references of sufficient background knowledge. Much qualified background knowledge comes from good experiments that have been done before. From a historical point of view, one needs to do experiments first and gain enough knowledge in order to do simulations properly. Experiments, on the contrary, do not necessarily need simulations' preparation. This view is challenged by Roush (2018) that it is a very weak claim, and the priority does not say privilege in a clear term. Suppose the results of the simulations are somehow more trustworthy. Should we say the epistemic privilege of experiments still stands, because the simulations require experiments to provide background knowledge to proceed? I believe it is arguable. As for Morgan (2005, 2012), epistemic privilege refers to the superiority in inferential power. This view is to some extent tailored to the Materiality Thesis that the material status plays the decisive role in justifying the inference to the target system. Such an account is too strong for every experiment must excel at justifying the inference to the target system compared to simulations. This is not always the case in practice. Another candidate in Morgan's account is the possibility of "confoundment" provided by experiments exclusively. To read it as a source of epistemic privilege, it is refuted by Parke (2014) that the confoundment in the experiments is not self-evidently good and will not imply any in-principle justificatory power that will make experiments better than simulations. However, I think this confoundment is a special value of the experiment, which I will elaborate in the following sections.

It seems that these criteria are not capable of communicating with each other and are also hopeless in forming a consensus. The reason this debate is unsettled is that they all own a distinctive explanation for the term "epistemic privilege". For Winsberg, it is the time sequence. For Morgan, it is inferential power. The above criteria are both to some extent faint because they keep a distance from the real scientific inquiry directed by humans.

Here I present a pragmatic perspective to discuss simulations and experiments' epistemic power and privilege. Instead of listing what experiments and simulations can and will achieve, and whether the way to achieve something can be compared, I consider if there is any kind of achievements that cannot be replaced by the other method. In other words, if there is any special knowledge generated by experiments or simulations which cannot be obtained by using the other method, and the special knowledge is of epistemic value and pragmatic worth, then none of the two methods can say it entitles epistemic privilege[2].

This criterion is pragmatic because it does not stop at the unsettled debate about the epistemic value of pure knowledge provided by simulations and experiments. Instead, it continues to

[2] This section by no means supports a radical and imprudent proposal to ground the epistemology of simulation entirely on a pragmatic basis. The epistemology of simulation should include much more discussions such as the way knowledge is formed and justified in technical details. My intention here is to use pragmatic perspective as a refute to the epistemic privilege claim.

consider whether and how acquiring that kind of knowledge is an interest of the focus of the researchers involved. The rationale of such consideration is to pay attention to the inquiry and the corresponding knowledge created, not only knowledge itself and the ways the knowledge is created. Such pragmatism is imported into social sciences to dedicates to answering and solve problems in the real world through generating useful knowledge and making scientific inquiries actual productive activities. Moreover, the attention toward the practical work of social scientists will raise an awareness that scientific research does not fit neat and objectivist criteria. Various disciplines work differently, local traditions culminate in distinct methodological practices which is unwarranted to talk in a unified fashion (Baert, 2011). In the next two sections, I shall present two kinds of special knowledge of pragmatic worth that gained through simulations and experiments respectively and cannot be replaced by the other one.

4.2 Simulation's Special Epistemic Power: Prediction

The special knowledge generated by simulations and cannot be replaced by experiments is the extensive predictive knowledge. Prediction is an accurate forecast of behaviors that have not yet been observed (Yarkoni&Westfall, 2017, p.1100). Simulations reach predictive knowledge in a considerably progressive manner by abstract modelling and computational power. Abstract modelling is the construction of the governing principles of simulations. The computational power enables a massive amount of executions of numerical treatments and provides us with the final results with higher and higher accuracy. If such construction of principles faithfully identifies the actual process running in the target system, then it is expected to provide an output which catches the thing that will happen in the target system correctly. Consider a case of pandemic simulation to see how a predictive output is generated. The number of people at different epidemiological stages is input. There is an amount of X population at the Stage A and an amount of Y population at the Stage B. Parameters are built to show the probability of changing epidemiological status from A to B. Other factors like transportation control or vaccination rate are also modelled in the mathematical system. Together running computation of the models will show the rate of change in the infectious status of the whole population. Such a simulation achieves empirical success in providing a predictive picture of the number of cases in a given country (Annas et al., 2020).

The prediction in social sciences is, in principle harder than in natural science for the reason that heterogeneity exists at the individual level (Squazzoni et al., 2013). So, a more intuitionistic and acquainted example of the prediction of simulations is the weather forecast simulation in natural science. The weather forecasting models are constructed to simulate the complex temperature structure in the atmosphere to predict the possible temperature of a given location on a short-term date[3]. Such simulations tend to give a reliable result and provide a basis for arguing the epistemic power of prediction (Parker, 2008a). The weather prediction is hard to imagine in a laboratory

[3] The progressive success of prediction by simulations may also benefit from the evolution of the models in a complicated technical manner. My intention here is only to point out the predictive power of simulations.

experiment with settings of same-kind objects. To know exactly what temperature range in a specific place at a particular time, creating a small world on an equal scale in a laboratory experiment and observation will not provide us with the same accurate result that simulations give.

There is a possible concern of "mere calculation". It acknowledges the empirical success of simulations yet doubts that simulations are just calculation tools that do not bring anything new at all. In this way, the epistemic value of simulations is in doubt to own a significance to be qualified to compare with the knowledge experiments would bring. To solve this concern, I shall first argue that simulations do have epistemic values and then explain why this epistemic value is special and not accessible to experiments.

First, mere calculation, i.e., running existing mathematical models and producing output, does have epistemic values simply because this process provides us with useful information which people do not yet know. Crudely put, the epistemic value is that simulations make implicit things explicit. It is crucial because, without advance and clarification, a lot of things will remain in a chaotic situation, and a confused grasp of the contents cannot be used by people efficiently to produce more knowledge (Hookway, 2004, p.120). For example, the formula of the difference of two squares is nothing more than the multiplication of polynomials. Yet the formula is of epistemic value because it simplifies the calculation process in practice. From a pragmatic perspective, making the calculations process more explicit is nontrivial and meaningful because it leverages people's understanding of complex things. The focal point in scientific research is purposeful human inquiry. Actions and results are assessed in light of practical consequences. This view indicates that not only do the fundamental laws and principles have epistemic value, but also information and knowledge originated from them. In the case of pandemic simulation, one could be perfectly aware of the movement of people and the epidemiology of virus, but the computing on these existing models provides us with the knowledge we want, i.e., the predictive infectious cases in the upcoming days. Even if the predictive case is imaginable, knowing it well with high credibility is different. Such articulation will be of great help for the clarification of scientific inquiry and avoid meaningless chaos.

Why is this epistemic power distinctively owned by simulations but not replaceable by experiments? After all, simulations can be regarded as application tools of principles and laws, it is possible that other kinds of scientific methods can also do it just well if they can input such rules. Simulations' computing power is the answer to this doubt. Simulation computation and human calculation provide qualitatively, not quantitatively different kinds of knowledge (Humphreys, 2004; Humphreys, 2009). It will be pretentious and implausible to say that other methods could recur the computation in the weather forecasting simulation step by step, simply because it is out of men's power. Experimental findings may provide us with principles and laws, but the intensive prediction has to be carried out by simulations. To make such extended prediction technically possible, experiments may need to translate or interpret the objects of interest in a computable manner, which will in turn make experiments essentially simulations.

4.3 Experiment's Special Epistemic Power: Confoundment

The special epistemic power entitled by experiments is the ability to find additional problems of interest. This is an illustration of Morgan's (2005) term "confoundment", which refers to the chance of observing new patterns, establishing new regularities, and therefore discovering new phenomena (p.326). While many parts of the confoundment argument are developing the view of experiments' epistemic power, they also fit the analysis structure from the pragmatic perspective well. Because the confoundment in social sciences encourages us to review our fundamental understandings of the world and can potentially form impacts socially.

First, we shall discuss how experiments create such confoundment and why simulations lack the ability to do so. The one thing simple yet important is that experiments are free from formal abstractions of the target system. In this way, there are two conditions for experiments to gain insights into the possible modifications to the current theories and assumptions. One is the irrelevant and unknown properties outside the scope of shared kindhood, and the other one is objects' freedom from formal abstraction to act with a degree of autonomy. In other words, the justified irrelevant factors have the freedom that enables their activities to break up the boundaries of the experimental settings potentially. It is the incompleteness of the kindhood and the authenticity in real scenarios provides the ground for the confoundment to emerge.

Why is confoundment absent in simulations? Because what confoundment refers to is the illuminations of something we still do not know about the interesting behaviour in the world, but not in the model. It is true that one may not be aware of the output in simulations when he starts, but the output is unknown, not unexpected. This is by no means denying the surprising moments in simulations. The point is that there is an invisible boundary constraining simulations to produce output out of the scope of parameters and models (Giere, 2009). The data obtained from the weather forecasting model is known within the limits of a variety of scientific principles. As for the experiments, since they are conducted in a real scene and rather than abstracted situations, and there are many other properties in the shared kindhood yet have not been identified or considered, the possibilities for experimental objects to step over the boundaries of assumptions and principles exist. The possibilities come from the degree of freedom for the shared kind of individuals, organisations, social interactions, and social artifacts to evolve and act outside of the boundaries of the experimental setting according to their potential undefined properties.

For example, when conducting experiments in classrooms and laboratories, economists run the risk of coming up with incorrect predictions or hypotheses about what will happen in the economy or having a totally inadequate understanding of the economic behaviour of the things they are studying. (Morgan, 2012, p.294). One may find out that the recognition of kindhood is inappropriate, other properties beyond the scope of kindhood matter. In other words, experiments have the capacity to push one to rethink the fundamental construction of models and simulations. This is an extension of the scope of the problem of scientific interest and opens up questions to the

standardised theories, background knowledge and so on. It can be read as of significant pragmatic value since it has the potential to rewrite the current principles, which will potentially reshape our understandings and reactions to the world, e.g., economic policies and corporation strategies, etc.

It might be true that the capacity of this sort of confoundment is not owned by experiments exclusively (Parke, 2014). However, this is indeed a distinctive way of extending the scientific investigation by questioning the very foundation of the research design, which cannot be replaced by simulations, thus suffice to refute the claim of epistemic privilege. In this way, experiments open up a broader scope of scientific inquiry which is of pragmatic value, while simulations answer some pressing also hard questions well, which is beyond the capability scope of experiments.

5 CONCLUSIONS

The comparison between simulations and experiments is of importance to both social scientists and philosophers of social sciences. For social scientists, they care about how simulations and experiments work in scientific practices. For philosophers of social sciences, they wonder that do these two methods work in the same way in producing knowledge. The abundant definition works together to draw comprehensive pictures of simulations and experiments for us to understand. Yet, in real world implications, most of them are mixed cases that both simulating and experimental traits can be found.

The Materiality Thesis is the most important account in dividing simulations and experiments. It can be analysed at two levels. The first level of analysis focused on the material status of the simulating and experimental systems in a straightforward but superficial way. It is challenged that simulations can be viewed as abstract systems since the realisation vehicle is not crucial and the material status is not meaningful. This level of analysis eventually fails because it cannot catch the nonmaterial part in social sciences experiments.

The other level of analysis focuses on the relationship between the simulating or experimental system to the target system. It is firstly attacked by a similar reason that the nonmaterial part in social sciences experiments also fails to make judgements on whether they are made of the same stuff. The Materiality Thesis overall is outdated and inaccurate in social sciences. More importantly, material similarity does not necessarily justify the construction of an experimental system or ensure the same material cause in the experimental process. What is important for an experiment to work well is the recognition that experimental objects share the same kindhood with the objects in the target system, thus achieving relevant similarity. Meanwhile, simulations are indeed abstract and formal, but they also aim at achieving relevant similarity with the target system but through good modelling.

Since simulations and experiments build very different systems to achieve relevant similarity with the target system, they will have different epistemic implications. Simulations offer predictive knowledge by their computing power which cannot be replaced by experiments. Experiments provide possibilities to find novel patterns which cannot be obtained by simulations. From a pragmatic perspective, because simulations and experiments are distinct in their offering, the claim of epistemic privilege is implausible.

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